

Differentiability

Derivatives are linear approximations

Recall if $f: \mathbb{R} \rightarrow \mathbb{R}$, we say that f is differentiable at x_0 if the limit

$$\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = L$$

exists. The limit is called the derivative denoted

$$L = \frac{d}{dx} f(x_0) = f'(x_0)$$

$\Rightarrow f$ is approximated by $f(x_0) + f'(x_0)(x - x_0)$ near x_0 , i.e.

$$f(x) - f(x_0) - f'(x_0)(x - x_0) = o(|x - x_0|)$$

Now consider $f: \mathbb{R}^n \rightarrow \mathbb{R}$. We can restrict f to a line passing through $x_0 \in \mathbb{R}^n$ and compute the derivative along the line. This is the directional derivative: Let $v \in \mathbb{R}^n$

$$D_v f(x_0) = \frac{d}{dt} f(x_0 + vt)$$

Directional derivative along elementary vectors e_i are called partial derivatives denoted

$$\frac{\partial}{\partial x_i} f(x_0) := \partial_{x_i} f(x_0) := D_{e_i} f(x_0)$$

for a general function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, a linear approximation near x_0 would look like

$$f(x) - f(x_0) + A(x - x_0)$$

where A is an $m \times n$ matrix

Def. Let $f: \Omega \rightarrow \mathbb{R}^m$, $\Omega \subset \mathbb{R}^n$, then f is differentiable at $x_0 \in \Omega$ if there exists a linear transformation $A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ s.t.

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0) - A(x - x_0)}{|x - x_0|} = 0$$

If A exists, A is called the total derivative, denoted $Df(x_0) = A$,

and it is given by the Jacobian

$$\text{Jac}_{x_0}(f) = \begin{pmatrix} \partial_{x_1} f_1 & \dots & \partial_{x_n} f_1 \\ \vdots & \ddots & \vdots \\ \partial_{x_1} f_m & \dots & \partial_{x_n} f_m \end{pmatrix}$$

Example: $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 + y^2$

$$Df(x, y) = (2x, 2y)$$

Non example: $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

All directional derivatives exist, but f is not differentiable

Prop. Let $f: \Omega \rightarrow \mathbb{R}$. If $\partial_{x_i} f$ exists and is continuous at x_0 . Then f is differentiable at x_0

Observe that $D_v f(x) = [Df(x)]v$

Define the gradient $\nabla f(x) = (Df(x))^T = \begin{pmatrix} \partial_{x_1} f(x) \\ \vdots \\ \partial_{x_n} f(x) \end{pmatrix}$

then $D_v f(x) = \nabla f(x) \cdot v$

Prop (Chain rule) $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $g: \mathbb{R}^m \rightarrow \mathbb{R}^p$ both C^1 ,
 then $g \circ f \in C^1$ and
 $[D(g \circ f)](x) = Dg(f(x)) \circ Df(x)$

Special case: change of variable, $\Omega, \tilde{\Omega} \subset \mathbb{R}^n$, $f \in C^1(\Omega)$

$\gamma \in C^1(\tilde{\Omega}, \Omega)$, c.o.v. given by $x = \gamma(y)$

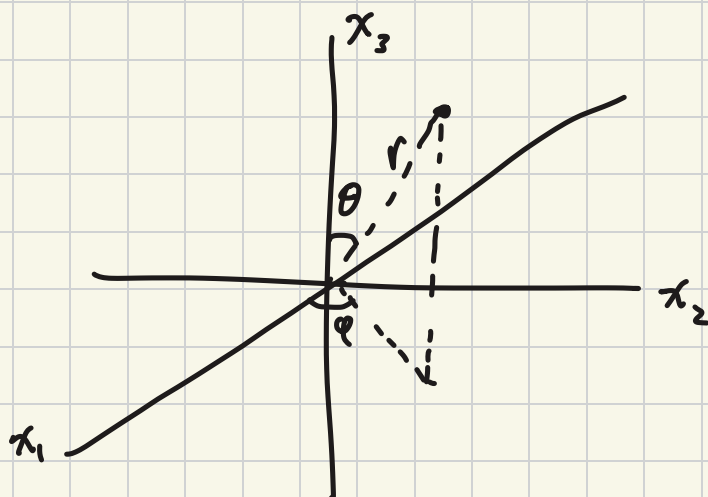
$$f \circ \gamma(y) = f(x(y))$$

$$[D(f \circ \gamma)](y) = (\partial_{x_1} f(x) \dots \partial_{x_n} f(x)) \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots \\ \frac{\partial x_2}{\partial y_1} & & \\ \vdots & & \end{pmatrix}$$

$$\leadsto \partial_{y_i} f(x(y)) = \frac{\partial x_1}{\partial y_i} \frac{\partial f}{\partial x_1}(x(y)) + \dots + \frac{\partial x_n}{\partial y_i} \frac{\partial f}{\partial x_n}(x(y))$$

Example: spherical coordinates

$$\gamma(r, \theta, \varphi) = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} r \sin \theta \cos \varphi \\ r \sin \theta \sin \varphi \\ r \cos \theta \end{pmatrix}$$



Some more notation: $\alpha \in \mathbb{N}_0^n$ is called a multiindex
" $(\alpha_1, \alpha_2, \dots, \alpha_n)$

$$\text{length: } |\alpha| = \sum_{i=1}^n \alpha_i$$

$$\text{for } x \in \mathbb{R}^n, \quad x^\alpha := x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$$

$$\Omega \subset \mathbb{R}^n, \quad f: \Omega \rightarrow \mathbb{R}, \quad \partial_x^\alpha f := \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_n}^{\alpha_n} f$$

$$\alpha! := \alpha_1! \alpha_2! \dots \alpha_n!$$

Def. The spaces of k -times continuously differentiable functions are denoted

$$C^k(\Omega) = \{f: \Omega \rightarrow \mathbb{R} \text{ s.t. } f^\alpha \in C^0(\Omega) \forall |\alpha| \leq k\}$$

$$C^k(\Omega; \mathbb{R}^m) = \{f = (f_1, \dots, f_m): \Omega \rightarrow \mathbb{R}^m \text{ s.t. } f_j \in C^k(\Omega)\}$$

Prop. C^k complete with norm

$$\|f\|_{C^k} = \max_{|\alpha| \leq k} \sup_{x \in \Omega} |\partial_x^\alpha f(x)|$$

Higher order approximation

Taylor's thm if $f \in C^k(\Omega)$, then

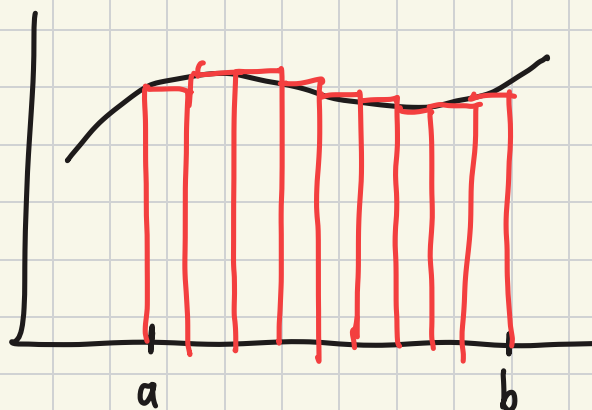
$$f(x) = \sum_{|\alpha| \leq k} \frac{\partial_x^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha + o(|x - x_0|^k)$$

Integration

Let $f \in C^0([a, b]; \mathbb{R})$. The Riemann integral of f is

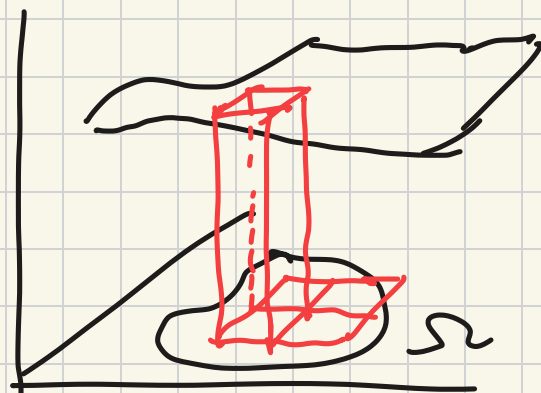
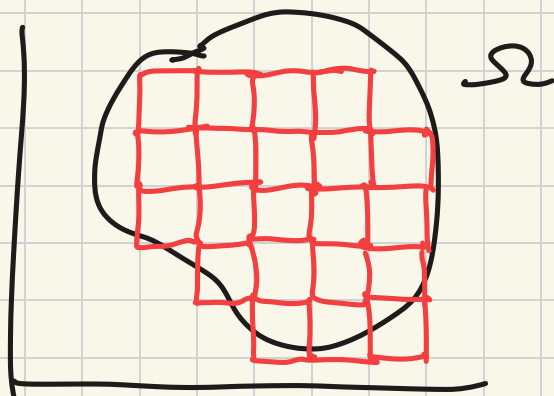
$$\int_a^b f(x) dx = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{\substack{j \in \mathbb{Z} \\ j/N \in [a, b]}} f\left(\frac{j}{N}\right)$$

Picture



More generally for $f \in C^0(\Omega; \mathbb{R})$, $\Omega \subset \mathbb{R}^n$,

$$\int_{\Omega} f(x) d^n x = \lim_{N \rightarrow \infty} \frac{1}{N^n} \sum_{\substack{j \in \mathbb{Z}^n \\ j/N \in \Omega}} f\left(\frac{j}{N}\right)$$



1-D warm up to vector calc + ODE

Fundamental thm of calc

Thm 1.) Let $f \in C^0([a, b]; \mathbb{R})$. Define

$$F(x) = \int_a^x f(t) dt$$

Then $F \in C^1([a, b])$ with $F'(x) = f(x)$

2) Let $F \in C^1([a, b])$. Define

$$f(x) = F'(x)$$

then
$$\int_a^b f(x) dx = F(b) - F(a).$$

Roughly: diff and integration are left and right inverses of each other

Local existence + uniqueness of ODEs

⑤
$$\begin{cases} \frac{du}{dt}(t) = F(t, u(t)) \\ u(0) = u_0 \end{cases}$$

Q. does there exist $u: (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^d$ that solves ⑤?

Q if so, is it unique?

Ex. $\frac{d}{dt} u(t) = u(t), \quad u(0) = 1$

Should expect $u(t) = e^t$ to be the unique sol.

Let's just consider the case $u: (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}$ for now

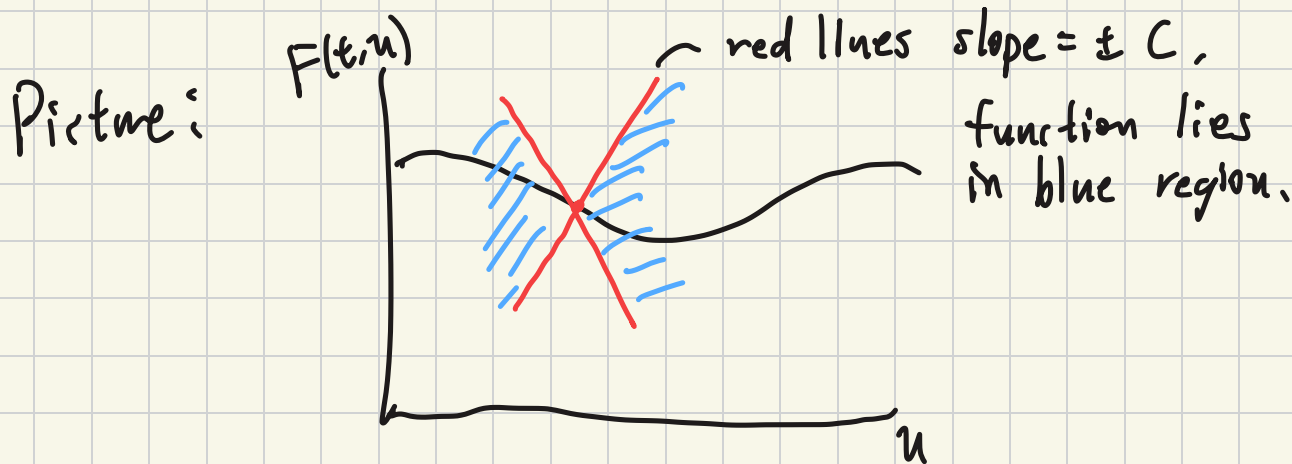
Thm (Picard-Lindelöf) Let $F: \mathbb{R}_t \times \mathbb{R}_u \rightarrow \mathbb{R}$ be a function that is continuous in t and Lipschitz continuous in u . Then $\exists \varepsilon > 0$ s.t.

$$\begin{cases} \frac{d}{dt} u(t) = F(t, u(t)) \\ u(0) = u_0 \end{cases}$$

has a unique solution for $t \in (-\varepsilon, \varepsilon)$.

Recall $F(t, u)$ is Lipschitz continuous in u if $\exists C > 0$ s.t.

$$|F(t, u_1) - F(t, u_2)| < C |u_1 - u_2|$$



Note that Lipschitz $\Rightarrow$ continuous. for any $\varepsilon > 0$, pick $\delta = \frac{\varepsilon}{C}$, then when $|u_1 - u_2| < \delta$,

$$|F(t, u_1) - F(t, u_2)| < C |u_1 - u_2| < C \delta = \varepsilon$$

Pf sketch. Let's try to solve the ODE using FTC.

$$\textcircled{*} \begin{cases} \frac{d}{dt} u(t) = F(t, u(t)) \\ u(0) = \mu \end{cases}$$

Using FTC, we find

$$u(t) = \mu + \int_0^t F(s, u(s)) ds.$$

It appears that we need to know u on the RHS in order to obtain u in this expression.

Idea: Put in a guess on the RHS and hope that the output gets closer to the solution.

Define operator $L: C^0([-\varepsilon, \varepsilon]) \rightarrow$ by

$$(Lw)(t) := \int_0^t F(s, w(s)) ds$$

Pick a random $w_1 \in C^0([-\varepsilon, \varepsilon])$. Let

$$w_2 = Lw_1$$

$$w_3 = Lw_2$$

$$w_4 = Lw_3$$

$\vdots$

Hope: the sequence $(w_j)_{j=1}^\infty$ in $C^0([-\varepsilon, \varepsilon])$ converges to the solution.

Note. $u \in C^0([-\varepsilon, \varepsilon])$ is a solution to $\textcircled{*}$ iff

$$Lu = u$$

Lemma. (Banach fixed point). Assume that L satisfies

$$\|Lw - Lv\|_{C^0} \leq \frac{1}{2} \|w - v\|_{C^0}$$

$\forall w, v \in C^0([-\varepsilon, \varepsilon])$. Then $\exists$ a unique $u \in C^0([-\varepsilon, \varepsilon])$ s.t. $Lu = u$.

Pf. Pick some $w_0 \in C^0$. define $w_j = L^j w_0$.

$$\text{Let } d = \|w_1 - w_0\|_{C^0}$$

$$\text{Observe } \|w_{j+1} - w_j\|_{C^0} \leq \frac{1}{2^j} d$$

Consider the infinite series

$$w_0 + \sum_{j=0}^{\infty} w_{j+1} - w_j$$

Observe that the partial sums are

$$w_0 + \sum_{j=0}^n w_{j+1} - w_j = w_{n+1}$$

and the series converges absolutely since

$$\|w_0\|_{C^0} + \sum_{j=0}^{\infty} \|w_{j+1} - w_j\|_{C^0} \leq \|w_0\|_{C^0} + \sum_{j=0}^{\infty} \frac{d}{2^j} < \infty$$

$$\Rightarrow \exists u \text{ s.t. } \lim_{n \rightarrow \infty} w_n = u$$

$$\Rightarrow Lu = \lim_{n \rightarrow \infty} Lw_n = \lim_{n \rightarrow \infty} w_{n+1} = u$$

uniqueness? Assume by contradiction $\exists u, \tilde{u} \in C^0$
s.t.

$$Lu = u \quad \text{and} \quad L\tilde{u} = \tilde{u}$$

Then $\|Lu - L\tilde{u}\|_{C^0} = \|u - \tilde{u}\|_{C^0}$
contradicting our contraction hypothesis.

End of pt of Picard Lindelöf

remains to check L is a contraction. Observe

$$\begin{aligned} |(Lw)(t) - (Lv)(t)| &= \left| \int_0^t F(s, w(s)) ds - \int_0^t F(s, v(s)) ds \right| \\ &\leq \int_0^{|t|} |F(s, w(s)) - F(s, v(s))| ds \\ &\leq \int_0^{|t|} C |w(s) - v(s)| ds \\ &\leq \int_0^{|t|} C \|w - v\|_{C^0} ds \\ &\leq C|t| \|w - v\|_{C^0} \end{aligned}$$

$\leadsto$ Choose $t \in (-\varepsilon, \varepsilon)$ with $\varepsilon < \frac{1}{2C}$

then $\sup_{t \in (-\varepsilon, \varepsilon)} |Lw(t) - Lv(t)| < \frac{1}{2} \|w - v\|_{C^0}$

$$\Rightarrow \|Lw - Lv\|_{C^0} \leq \frac{1}{2} \|w - v\|_{C^0}$$

